Diseases of Crucifers

1. **Damping off and Wire stem** - ***Pythium aphanidermatum*** ***Rhizoctonia solani***
2. **Club root** - ***Plasmodiophora brassicae***
3. **White rust** - ***Albugo candida* (syn:*Cystopus*)**
4. **Powdery mildew** - ***Erysiphe polygoni***
5. **Anthracnose** - ***Colletotrichum higginisianum***
6. **Downy mildew** - ***Peronospora parasitica***
7. **Alternaria leaf spot or black spot** - ***Alternaria brassicae, A. raphani, A. brassicicola***
8. **Black leg** - ***Leptosphaeria maculans (IS- Phoma lingam)***
9. **Cabbage yellows** - ***Fusarium oxysporum* f sp. *conglutinans***
10. **Black rot** - ***Xanthomonas campestris* pv. *Campestris***
11. **Mosaic** - ***Cauliflower mosaic virus* (CaMV)**

Damping off and Wire stem - *Pythium aphanidermatum, Rhizoctonia solani*

Wire-stem symptom

Rhizoctonia damping off, blight, and rot

Damping off and Wire stem *aphanidermatum, Rhizoctonia solani*

Wire stem

Damping off and Wire stem - *Pythium aphanidermatum, Rhizoctonia solani*

Crown rot symptom

Damping off and Wire stem - *Pythium aphanidermatum, Rhizoctonia solani*

More serious in nursery beds as pre emergence and post emergence damping off

It is also a **serious problem after transplanting** in the main field

Symptoms

- First sign as **reddish brown discolouration on the stem** at the soil level extends from 1 to 2 cms below
- The area is **constricted followed by girdling turns woody**
- Plant may be **bent or twisted without breaking**
- More in sandy soils than in clay soils

Damping off and Wire stem - *Pythium aphanidermatum, Rhizoctonia solani*

Symptoms contd…

- In older plants it causes **bottom rot and head rot** also in cabbage in which lower leaves droop, dry out turn dark
- Infection may progress to upper leaves and eventually reaches the head causes **head rot in cabbage and cauliflower**
- In **root crops the fungus advances** into the crown of fleshy root tissues turn brown causing head rot

Damping off and Wire stem - *Pythium aphanidermatum, Rhizoctonia solani*

Disease cycle

- Survives as mycelium and sclerotial bodies in soil for indefinite periods
- Spreads through rain and irrigation water

Damping off and Wire stem - *Pythium aphanidermatum, Rhizoctonia solani*

Management

- Changing the nursery site
- Nursery bed should be raised in soils with **near neutral pH with good drainage**
- Deep ploughing
- Disinfection of beds – soil solarisation/ fungicides
- Soil amendment with neem cake @1kg/m^2

Damping off and Wire stem - *Pythium aphanidermatum, Rhizoctonia solani*

Management contd……

- Growing resistant varieties
- Selection of healthy seed
- Seed treatment with biocontrol/ fungicides/mycorrhiza
- **Shallow sowing for rapid seedling emergence**
- Biological control

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Club root of crucifers or Finger and toe disease
Plasmodiophora brassicae

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Important disease in **temperate climates**

A very serious disease of cauliflower, cabbage, broccoli and Chinese cabbage and in India **common only in Darjeeling**.

Symptoms

- Infected plants have pale green to **yellowish leaves** initially.
- Infected plants - **wilting in the middle of hot, sunny days, recovering during the night**
- Primary, secondary and tertiary roots are **malformed into thick, fleshy growth**
- **Basal portion of the root is mostly normal** and tips are malformed

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Symptoms contd…

- Characteristic symptoms become apparent in advanced stage of root infection as **spindle like, spherical, knobby, or club-shaped swellings.**
- The swellings may be few, or they coalesce and cover the entire root system
- Hypertrophy causes **malfunctioning of xylem vessels** followed by drooping of leaves
- Stunting - reduction in size of heads
- In advanced stages the affected tissues are invaded by **saprophytes mostly soft rot bacteria** which turns black in colour

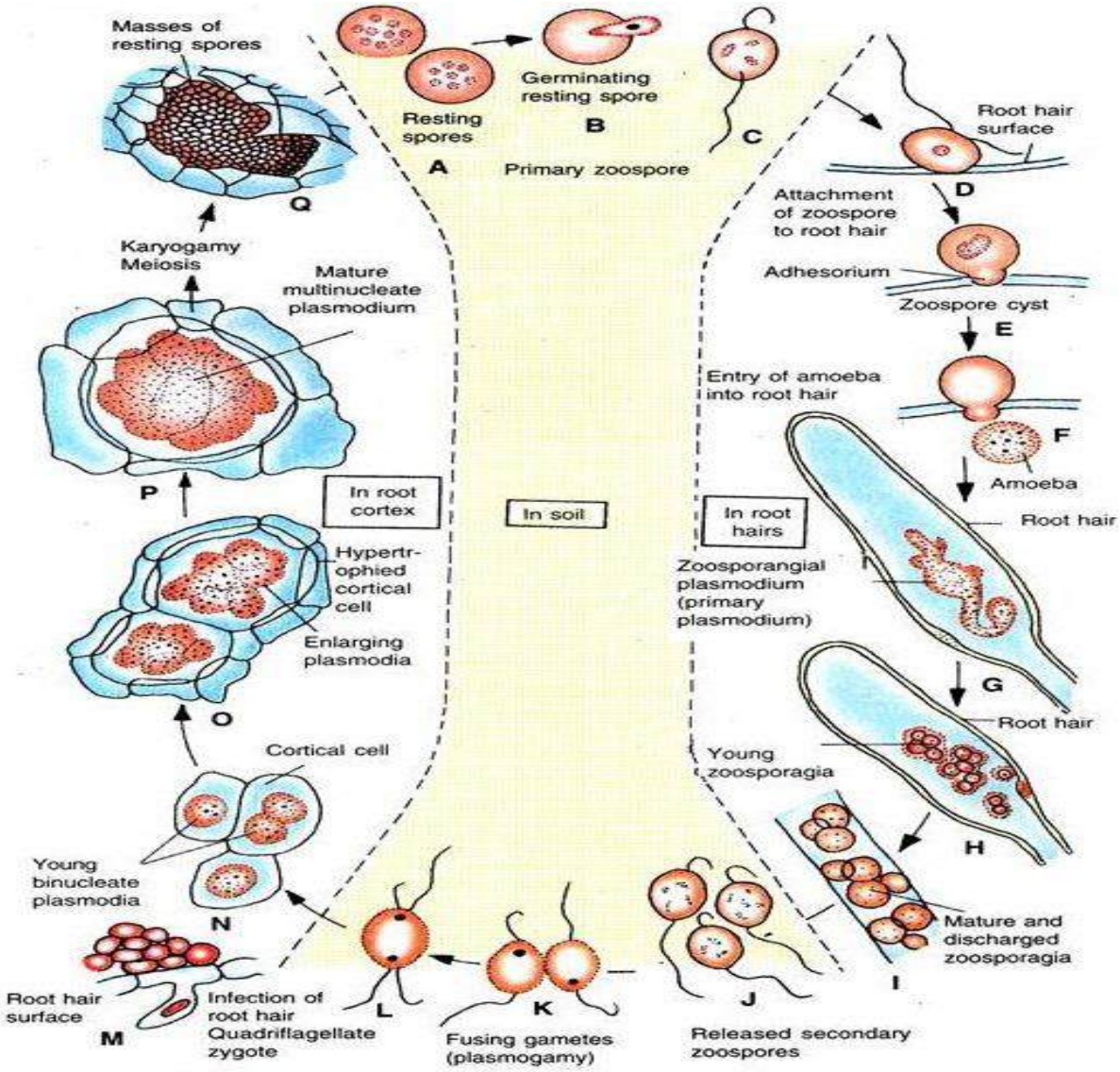
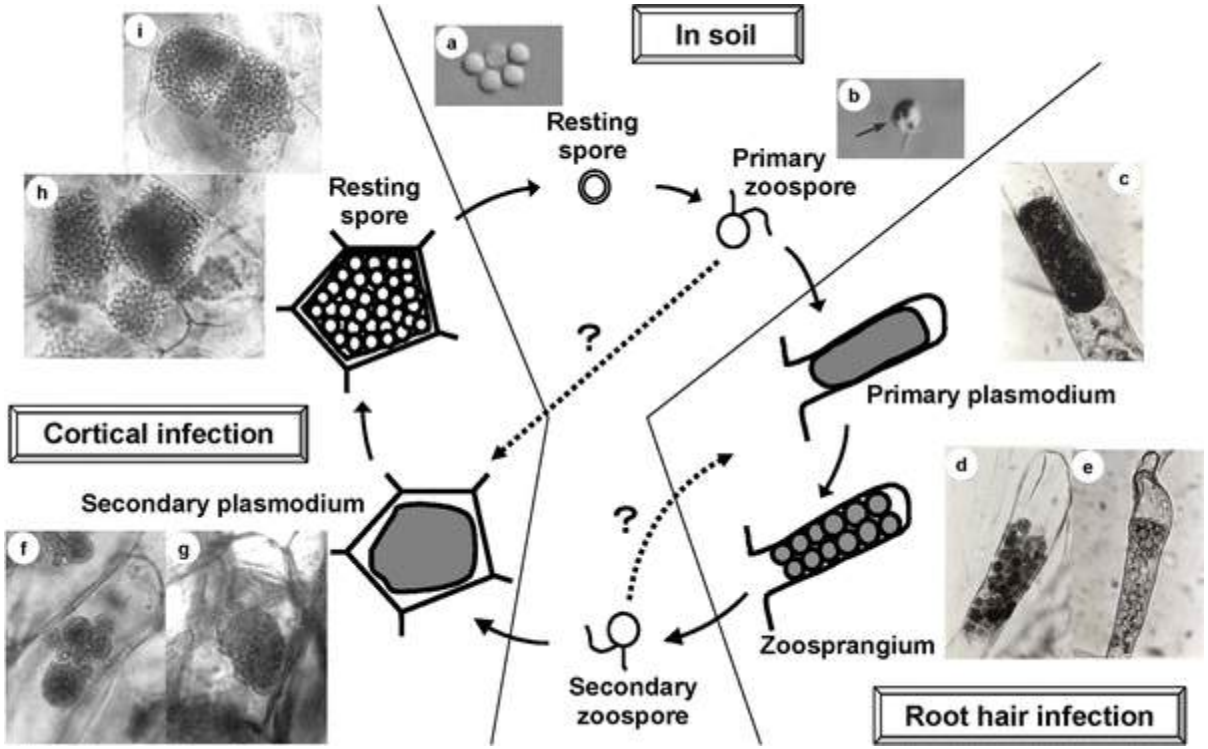


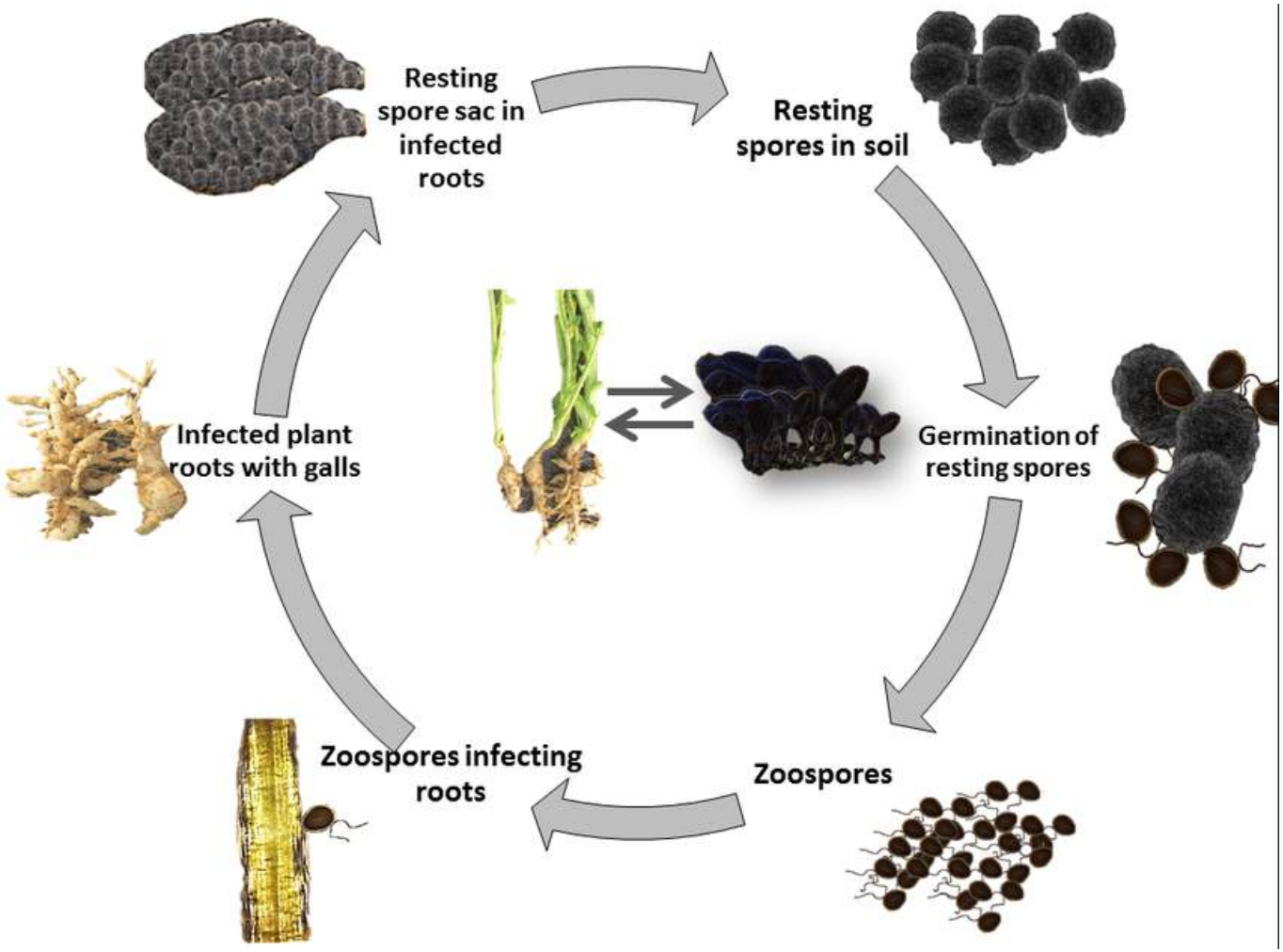
Fig. 5.8. Pictorial diagram of the life cycle of *Plasmodiophora brassicae*.

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*



Disease cycle

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*



Disease cycle

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Pathogen

- **Naked plasmodial thallus, holocarpic**, with plasmodial type of feeding
- **Somatic phase is a plasmodium which occurs inside the host cells .**
- **Obligate endoparasites**
- Parasitize plants by **Hypertrophy** (Making enlargement)& **Hyperplasia** (Abnormal multiplication of cells).

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Disease cycle and epidemiology:

- The fungus can survive in soil for several years in the form of **resting spores** which are disseminated by water and other cultural operations.
- Seedlings raised in infested soil also help in spreading of the pathogen to uninfested fields.
- The fungus can also **perennates on wild cruciferous plants.**
- The pathogen enters the host through the **root hairs and injured roots.**
- After entry, the invaded host cells enlarge and the fungus multiplies to develop into plasmodium.

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Disease cycle and epidemiology contd…

- As a result of cell division, the fungal spores are distributed to the newly formed cells which give the club shaped appearance to the root tissues.
- Such infected roots decay and release the spores in the soil.
- Soil temperature ranging from 9-30° C, moisture and pH play an important role in disease development and its progress.
- **Cool and wet climate along with acidic soils (pH 5.0-6.3)** are most favourable for development of the disease.

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Management

- Use of seedlings from disease free fields
- Avoid excess irrigation
- Plant cabbage and other susceptible cruciferous crops in **well drained fields that have a pH of 7.2**
- **Crop rotation for 12 years,** it persists in soil for 10yrs
- **Add hydrated lime to soil to increase pH to 7.2** (6 weeks before planting **@ 2.5t/ha**)

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Management contd…

- Soil drenching or dipping of the seedlings in benomyl (0.05%) or carbendazim (0.05%) solution for 15-20 minutes before transplanting helps in reducing the disease incidence.

- Besides these, **combination of calcium cyanamide with nitrogen and boron completely inhibits clubbing and root hair** invasion by suppressing the resting spore germination.

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Management contd…

- Soil amendment with **oyster shell powder + mineral ash** reduces the disease incidence
- Sowing of **rape seed mustard as bait crop 4- 5 weeks** before planting of cabbage and braccoli
- Use resistant cultivars braccoli- OR1

  raddish- Sexafire

Club root of crucifers or Finger and toe disease: *Plasmodiophora brassicae*

Management contd…

- Biocontrol agents like ***Pseudomonas fluorescens* and *Streptomyces graminofaciens*** helps in reducing size and growth of club as well as root infection
- **Treat the soil of seed bed areas** formaldehyde 4%

White rust - *Albugo candida*

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White rust - *Albugo candida*

Stag head

White rust - *Albugo candida*

It is the **most common disease of radish** than other crucifers and cause considerable losses when it occurs along with **downy mildew.**

Cauliflower, cabbage, brussel sprout - less susceptible

Mustard, turnip and raddish - **more susceptible**

Symptoms

- Two types of infection *i. e.*. **local or leaf phase** and the **systemic or stag head phase** occur in different crucifers.
- Local infection results in isolated pustules on the lower surfaces of leaves, stems,flowers. which are raised, glossy white blisters 1 to 2 mm in diameter.
- Pustules coalesce to **form irregular patches**

White rust - *Albugo candida*

Symptoms contd

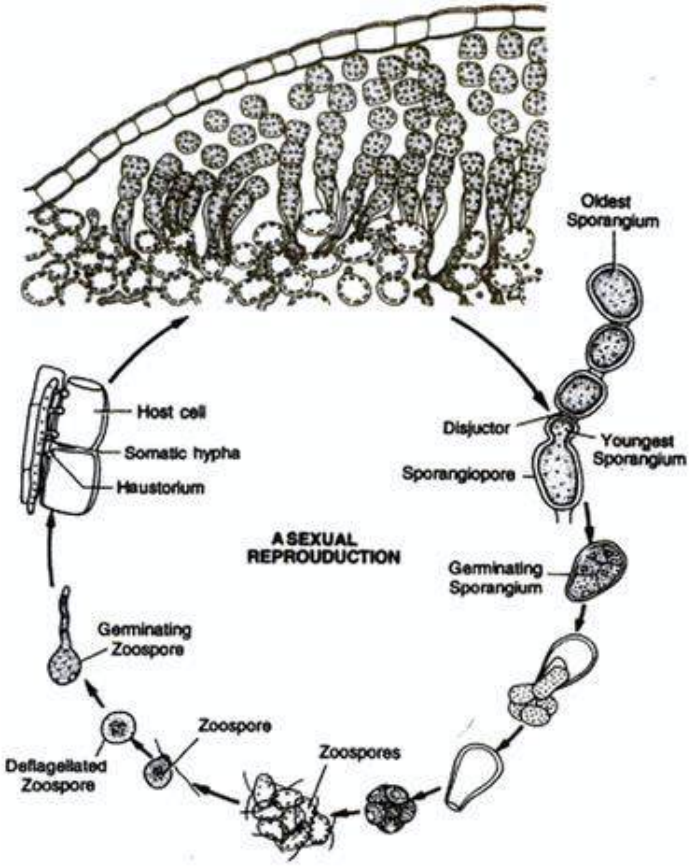
- The **inflorescence may become enormously thickened** (12-15 times than the normal thickness) known as **stag head** twisted in zig zag manner, fleshy and greenish leading to sterility.

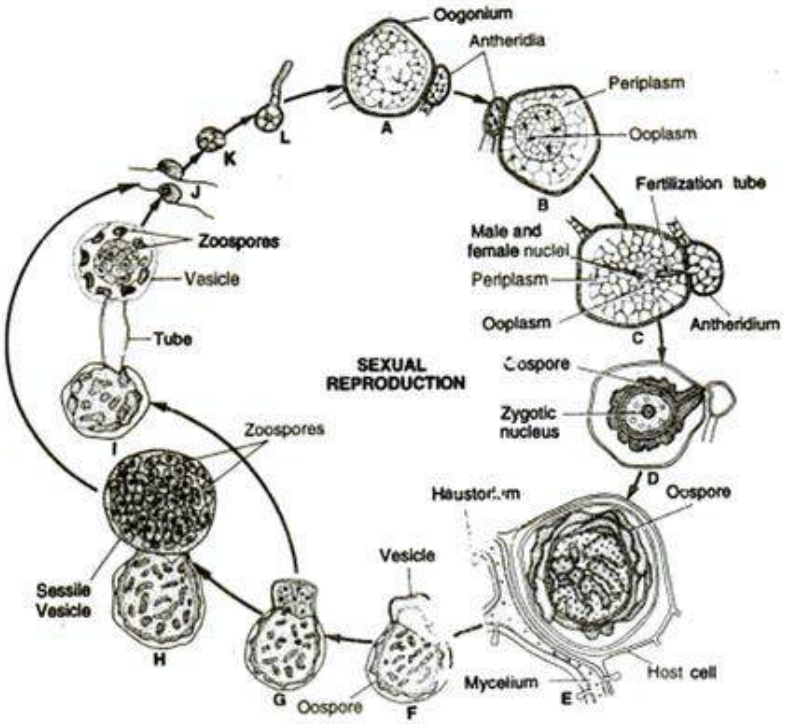
- Systemic infection stimulates **hypertrophy and hyperplasia** resulting in enlarged and **distortion of the floral parts including petals, pistils and anthers.**

- The **epidermis ruptures exposing white spore mass** which gives the pustule powdery appearance

- **Plants malformed** beyond recognition

White rust - *Albugo candida*





White rust - *Albugo candida*

Pathogen

- It is an **obligate parasite reproducing** both sexually and asexually.
- Mycelium is **intercellular and non-septate**.
- Numerous short **sporangiophores arise from the mycelium, which are arranged in rows** beneath the host epidermal layer press on the epidermis to cause the pustule/ blister.
- Inside the pustules, **sporangia are produced in chains** arising from the sporangiophores at the base of the cavity.

White rust - *Albugo candida*

Pathogen

- The **sporangia are hyaline, spherical, thick walled, hyaline** and interconnected **by a pad of gelatinous disc like tissue called disjunctor.**

- In the presence of moisture, these pads swell and disintegrate, freeing the sporangia in the chains and they germinate either by germ tube or by producing zoospores.

- The oospore is spherical with a thick irregular wall, deep yellow to dark brown in colour.

White rust - *Albugo candida*

Disease cycle and epidemiology:

- The fungus survives either as oospores in the infected plant debris or **mixed with seeds or as perennial mycelium in some weed hosts**
- These oospores may germinate in the ensuing crop season to produce zoospores, which serve as primary inoculum.
- Secondary spread is through sporangia produced on the spots caused by primary inoculum.
- Moisture is essential for germination of sporangia through zoospores.
- The optimum temperature for germination of sporangia by zoospores is 20° C.

White rust - *Albugo candida*

Management

- Sanitary measures
- Crop rotation
- **Destruction of weeds**
- Maintaining proper spacing
- **Avoid excess application of nitrogen**
- Spray 0.8% B.M or any copper fungicide
- Spraying with the initiation of the disease, with metalaxyl + mancozeb (0.25%) followed by sprays of Bordeaux mixture (4:4:50) or mancozeb (0.25%) or copper oxychloride (0.30%) and repeat at 10 to 14 days interval.

Powdery mildew - *Erysiphe polygoni*

Powdery mildew – *Erysiphe polygoni*

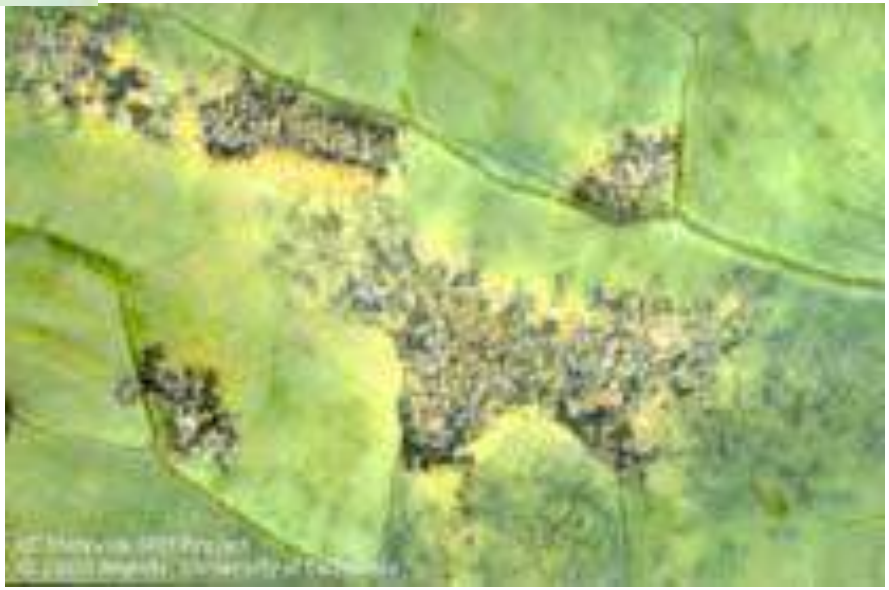
- It is a common disease in crucifers, *i.e.*, Cabbage and cauliflower
- It has been reported from several parts of the world.
- It occurs in late summer and autumn.
- Moderate to **dry-warm weather and low relative humidity** proves to be favorable for the growth of the pathogen.

Powdery mildew – *Erysiphe polygoni*

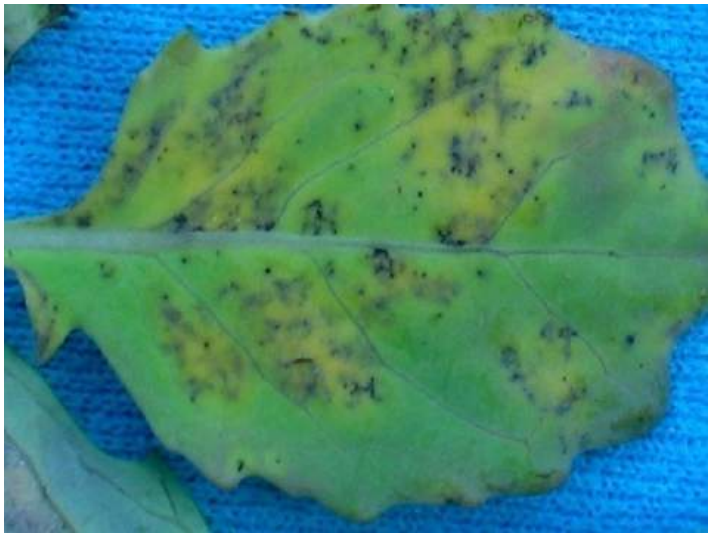
Management

- Application of Dinocap@0.2%

Downy Mildew – *Peronospora parasitica*



Downy Mildew – *Peronospora parasitica*



Downy Mildew – *Peronospora parasitica*

Raddish, cabbage, cauliflower, mustard, and knol-khol.

Symptoms

- The first symptoms of the disease are evident as **small brown spots on the lower surfaces** of the leaves.
- The spots are covered gradually with the **downy growth** consisting of sporangiophores and sporangia of the fungus
- **Upper surface of the leaves corresponding to spots** turns light yellowish in colour

Downy Mildew – *Peronospora parasitica*

Symptoms contd…

- Young leaves having a number of such spots turn **yellow and drop off** early whereas older leaves usually persist.
- **In the inflorescence: flower stalk becomes thick, elongated and curved.**
- **On curds of cauliflower and broccoli** as well as **on heads of cabbage** black sunken spots are produced.

Downy Mildew – *Peronospora parasitica*

Pathogen:

The fungus responsible for this disease is *Peronospora parasitica* (Pers.) Fr. which is an obligate parasite.

- The mycelium of the fungus is strictly intercellular with large, finger-shaped intracellular haustoria which become clavate and branched and nearly fill the cell cavity.
- The sporangiophores are produced during darkness, which are divided into primary and secondary branches (dichotomously branched), which ultimately bifurcate to form sterigmata which bear the single sporangium at the tip.

Downy Mildew – *Peronospora parasitica*

Pathogen:

- The **sporangia are broadly oval, ellipsoidal** and hyaline and they fall off mainly by hygroscopic twisting of the sporangiophores in response to change in atmospheric humidity.
- The **oospores are globose** measuring 26 to 43 µm in diameter.
- They are enclosed in crest like folds and appear pale yellow in colour and germinate by germ tube.

Downy Mildew – *Peronospora parasitica*

Disease cycle and epidemiology

- The downy mildew pathogen perennates as oospores in senescence host tissues
- It also survives as **conidia on leaves and inflorescence and as latent systemic mycelium in seeds or infected plant debris.**
- Infections are favoured by **low temperatures and high atmospheric humidity following rain or dew**.
- The penetration is usually direct but occasionally also occurs through stomata.

Downy Mildew – *Peronospora parasitica*

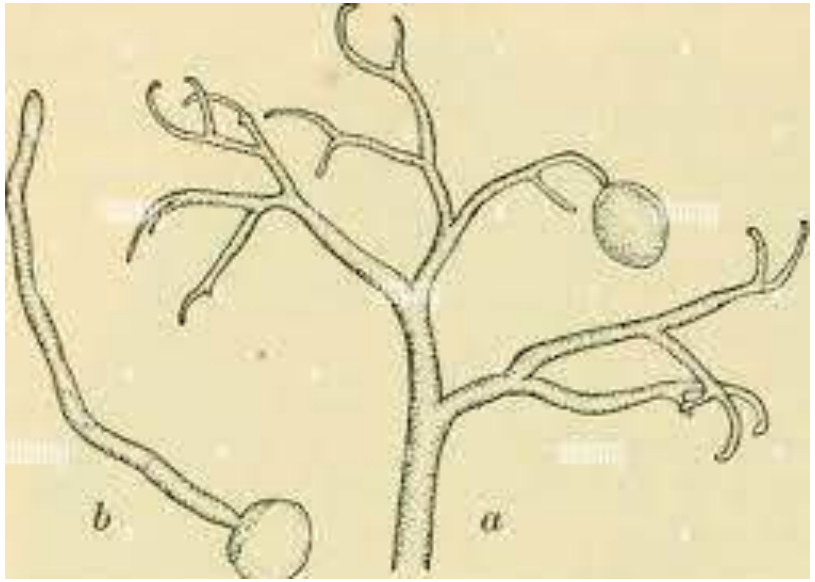
Disease cycle and epidemiology contd..

- Primary infection occurs due to soil borne oospores while zoospores released from sporangia found on the cotyledons or hypocotyls' favour secondary spread.

- Water droplets also help the pathogen dispersal over short distances.

- The optimum temperature for conidial germination is **8 - 12 ^{o} C** while penetration of the host by infection hyphae and formation of the haustoria occurred most rapidly at **16 ^{o} and 20-24 ^{o} C**, respectively.

Sporangiophore, Sporangia,



Downy Mildew – *Peronospora parasitica*

Management

- Collect and destroy the infected plant debris and perennial weed hosts.
- Crop rotation with **non cruciferous plants**.
- Use disease free seeds, seed treatment with metalaxyl + mancozeb (@0.3%)
- **Hot water treatment of seeds at 43 - 50°C for 20 minutes** and is also effective.
- Spray the crop with metalaxyl + mancozeb (0.25%) or mancozeb (0.25%) or copper oxychloride (0.3%) and repeat at 10 to 14 days interval.

Alternaria leaf spot: *Alternaria brassicola, A.brassicae, A.raphani*

Alternaria leaf spot: *Alternaria brassicola, A.brassicae, A.raphani*

Alternaria leaf spot: *Alternaria brassicola, A.brassicae, A.raphani*

Brown spots on the cauliflower head Alternaria leaf spot.

Alternaria leaf spot: *Alternaria brassicicola, A.brassicae, A.raphani*

Alternaria diseases are the most common and serious menace to the cultivation of crucifer vegetables world over.

Infection on **young leaves, stems and siliquae** generally results in heavy yield losses as well as **quality of seed**.

Symptoms:

- The leaf spots incited by ***A. brassicicola*** **appear as minute dark brown to black spots, which may enlarge by forming concentric rings** and surrounded by a yellow halo of chlorotic tissues.

- In humid weather, the fungus appears as **a bluish growth in the centre of the spots.**

Alternaria leaf spot: *Alternaria brassicicola, A.brassicae, A.raphani*

Symptoms contd…

- The spots by ***A. brassicae*** are similar to *A. brassicicola,* but smaller ***i.e., < 1 cm and light brown*** in colour
- The spots caused by ***A .raphani* are slightly yellowish**.
- Browning of cauliflower/ broccoli heads occur which generally begin at the margin of the individual flower or flower cluster.
- Plants grown from **infected seeds show dark necrotic elongated lesions on the main axis, inflorescence, branches and on the siliquae.**

Alternaria leaf spot: *Alternaria brassicicola, A.brassicae, A.raphani*

Symptoms contd…

- Infected seeds remain small in size, shrunken, discoloured or covered with fungal growth and have low viability.
- **Turnip and rutabaga foliage and roots** may also become infected in field **but root symptoms develop after being placed in storage**

Alternaria leaf spot: *Alternaria brassicicola, A.brassicae, A.raphani*

Symptoms contd…

- The leaf spots are nearly circular, often zonate and are of various shades of brown to black.
- **Radish plants kept for seed purpose are severely affected by *A. raphani*.**
- On leaf, the spots are raised, spherical to elliptical and 0.5-1 cm in diameter, black sporulation may also appear on the lesions in the humid weather, the center soon dries and may drop down.

Alternaria leaf spot: *Alternaria brassicicola, A.brassicae, A.raphani*

Pathogen(s):

Three species of *Alternaria* are *A. brassicicola* (Schw.) Wiltshire, *A. brassicae* (Berk.) Sacc.and *A. raphani* Groves

Alternaria leaf spot: *Alternaria brassicicola, A.brassicae, A.raphani*

Disease cycle and epidemiology:

- The pathogen perennates on infected plant debris, in or on infected seeds and cruciferous weeds.
- Moderate temperature (**21-27ºC**), high relative humidity (95-100 % at least for 18h) and splashing rains favour the disease development and spread.

Alternaria leaf spot: *Alternaria brassicicola, A.brassicae, A.raphani*

Disease cycle and epidemiology contd…

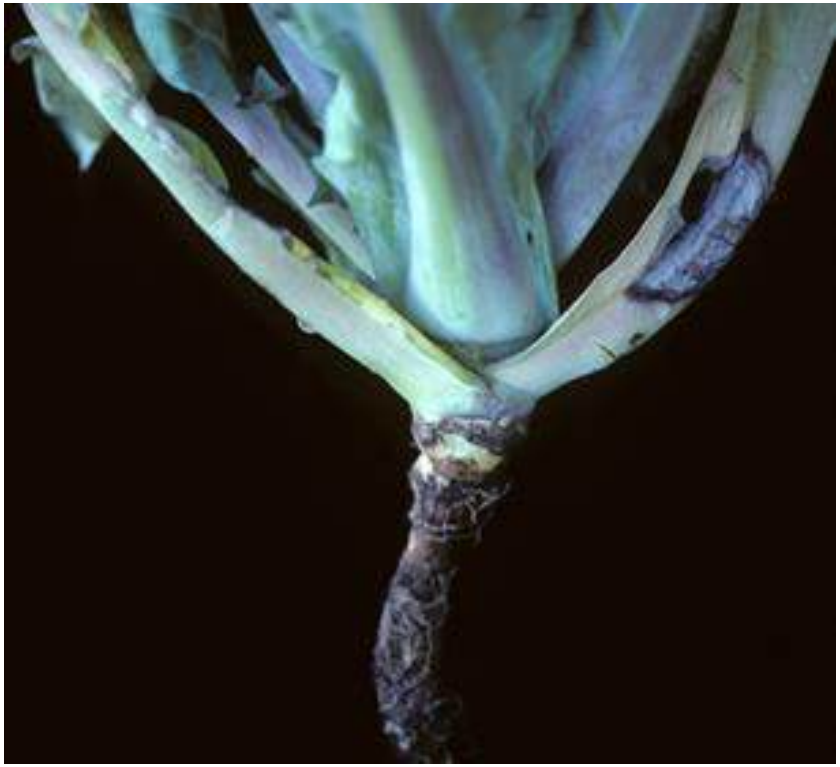
- The optimum temperature for sporulation is 18-24 and 20-30°C for *A. brassicae* and *A. brassicicola*, respectively, at which both fungi produce spores in 12-14 h.
- At temperature >24°C, sporulation of *A. brassicae* is inhibited.
- *A. raphani* infection progresses rapidly at 22-26°C.

Alternaria leaf spot: *Alternaria brassicicola, A.brassicae, A.raphani*

Management

- Collect and destroy the infected plant debris. Use disease free seeds and treat them with captan (0.3%).
- **Hot water treatment at 50^0c** / 30min
- Seed treatment with agrosan
- Spray the crop with mancozeb (0.25%) or copper oxychloride (0.3%) and repeat at 10 to 14 days interval.

Blackleg - *Phoma lingum* (PS: *Leptosphaeria maculans*)



Blackleg - *Phoma lingum* (PS: *Leptosphaeria maculans*)

Blackleg - *Phoma lingum* (PS: *Leptosphaeria maculans*)

- Black leg also known as **dry rot**
- It is an **extremely destructive disease of crucifers** worldwide.
- The fungus **infect all parts of the plant, including stem and leaves.**
- Stem damage lead to **cankering and severing** of the plant at the base.
- It is the most serious symptom of the disease which causes **plant lodging and yield loss**

Blackleg - *Phoma lingum* (PS: *Leptosphaeria maculans*)

Symptoms

- **Lesions exude a rose-pink** spore mass during moist conditions.
- On leaves, black leg appears as small **light brown lesions with a purplish outline.**
- The **lesions are linear** and irregularly distributed on the leaves.
- On **stems and bulbs,** it appears as **large brown-black dry and cracked lesions.**
- As the disease develop further the **bulbs crumple, leaves die and finally fall off** and stems severe at the point of the lesion.

Blackleg - *Phoma lingum* (PS: *Leptosphaeria maculans*)

Favourable conditions

Wet and windy conditions favor the development of the disease.

Disease cycle

Infected canola stubble serves as important source of inoculum of *L. maculans* which bears pseudothecia from which ascospores are released.

Besides the left over debris, also carried over by seeds which further get spread by irrigation, rainwater splash, wind, farm machinery and equipment dispersing the spores from lesions.

Blackleg - *Phoma lingum* (PS: *Leptosphaeria maculans*)

Management

- Before sowing **field sanitation of infected crop debris** should be ensured for good cultivation.

- **Certified and disease free seeds** should be used for sowing.
- **Crop rotations with non-host crops** for 3-5 yrs showed decrease in trend of infection
- Use of *L. maculans* tolerant cultivars also reduces the prevalence of black leg in areas affected from severe infections.

Blackleg - *Phoma lingum* (PS: *Leptosphaeria maculans*)

Management contd..

- **Brassinin a vital plant defense product**, produced by crucifers **is detoxified by the fungus** to indole-3-carboxaldehyde using brassinin oxidase.
- Synthesis and characterization of detoxification inhibitors, which shows cytotoxic as well as antifungal activity, can be used in combating the phytopathogenic fungus *L. maculans*

Fusarium yellows of cabbage
Fusarium oxysporum f.sp. *conglutinans*

Fusarium yellows of cabbage
Fusarium oxysporum f.sp. *conglutinans*

Fusarium yellows of cabbage
Fusarium oxysporum f.sp. *conglutinans*

This is vascular fungal disease, **common when the crop is grown in warm condition**

Symptoms

- Normally symptoms are well marked **2-4 weeks** after transplanting
- Affects the plant at any age.
- **Lifeless yellowish green colour of foliage** is the first sign of the disease.

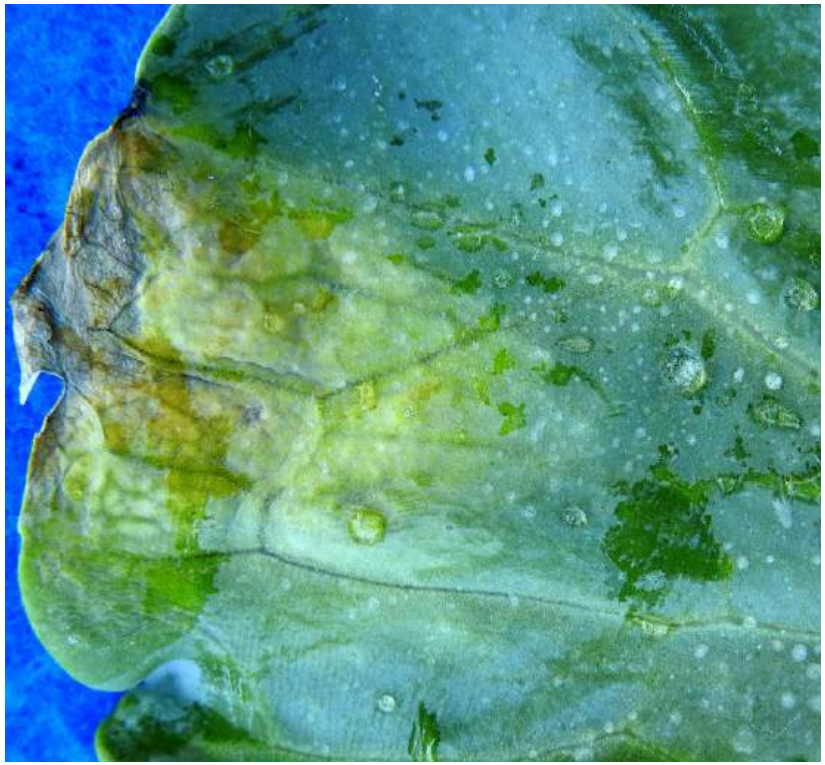
Fusarium yellows of cabbage
Fusarium oxysporum f.sp. *conglutinans*

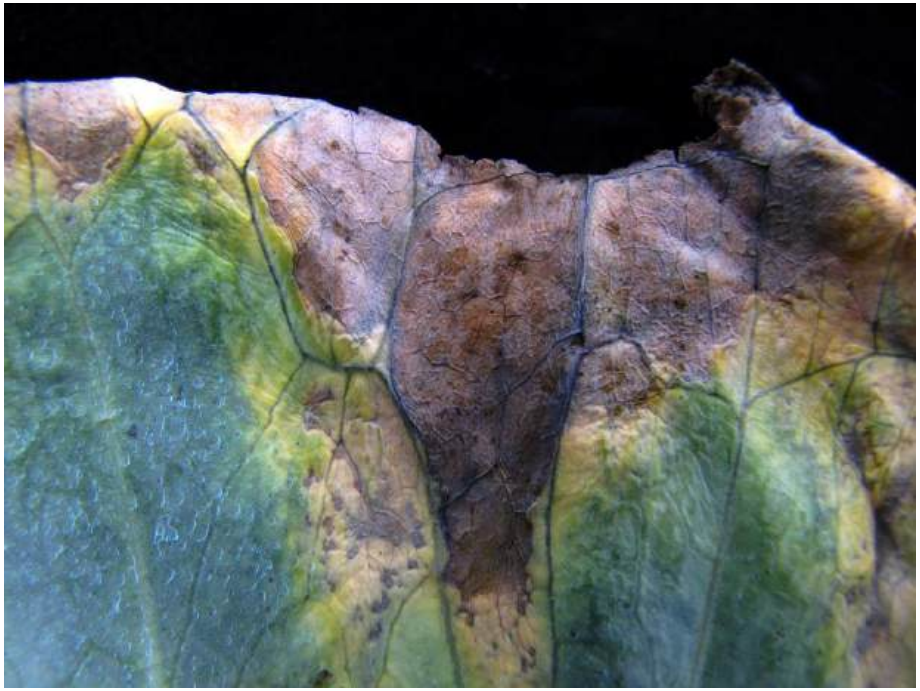
- Lower leaves yellow first & then spread to upper leaves.
- Yellowed tissues turns brown, dies become brittle
- Curling of stems & leaves is also observed.

Management

- Crop rotation & other cultural methods.
- Avoid planting of diseased seedlings carrying infected soil into diseased free areas

Black rot - *Xanthomonas campestris* pv. *campestris*





Black rot - *Xanthomonas campestris* pv. *campestris*

Black rot - *Xanthomonas campestris* pv. *campestris*

Symptoms

- Starts -leaf margins as chlorotic or yellow (angular) areas
- The yellow area extends to veins & mid rib forming - ‘v’ shaped chlorotic spots
- Veins and veinlets turn brown and finally black
- The vascular blackening extend beyond affected veins to midrib, petiole and stem

Black rot - *Xanthomonas campestris* pv. *campestris*

Symptoms contd…

- In advanced stages, infection may reach the roots system and blackening of vascular bundles occur
- Bacterial ooze can also be seen on affected parts
- If the infection is early, the plants wilt and die
- If the infection is late plant succumb to soft rot &die

Black rot - *Xanthomonas campestris* pv. *campestris*

Pathogen:

- It is a small, rod shaped, aerobic, gram negative, non-spore forming bacterium.

- The bacterium has a single polar flagellum (monotrichos) and it is catalase positive, hydrogen sulphide positive, oxidase negative and does not produce nitrate or indole.

Black rot - *Xanthomonas campestris* pv. *campestris*

Pathogen:

- It produces a yellowish extracellular polysaccharide (EPS) called Xanthan on media containing glucose.
- *X. c. pv. campestris* hydrolyses starch, a characteristic that is used for easy recognition of the bacterium on Schaad‘s selective medium.
- Its growth is inhibited or retarded in acidic range of pH.

Black rot - *Xanthomonas campestris* pv. *campestris*

Disease cycle and epidemiology:

- The pathogen survives in infected seeds, diseased plant debris and on cruciferous weeds
- When the infected seed germinates, the seed coat that comes out along with the cotyledons serves as the source of primary inoculum.
- The bacterium enters cotyledons through the marginal stomata and passes into true leaves through hydathodes.
- On entry, the bacteria move to the xylem vessels where they multiply rapidly and move up and down in the plant.

Black rot - *Xanthomonas campestris* pv. *campestris*

Disease cycle and epidemiology contd..

- When the disease inoculum is present in the field, secondary spread is facilitated by splashing of contaminated soil by irrigation water and/ or rains, blowing of fallen leaves and occasionally by insects.

- The optimum temperature for the disease development is about 26.5 to 30°C, minimum being 5°C and maximum 36°C.

- Heavy rains have been reported to be responsible for the fast spread of the pathogen and the disease through splashes.

Black rot - *Xanthomonas campestris* pv. *campestris*

Management

- Seed treatment with Hg Cl_2 solution for 30 min or Agrimycin or Aureomycin 0.01%
- Hot water treatment at 50^0c / 30min, for killing seed borne inoculum followed by a 30min dip in streptocycline 100ppm
- Spray Agrimycin-100 or Streptocycline-50ppm at transplanting, curd formation and pod formation

Black rot - *Xanthomonas campestris* pv. *campestris*

Management contd…

- Crop rotation for 2-3 yrs with non cruciferous crop
- Drenching seed bed with 5% formalin or any antibiotic solution in nursery beds

Resistant varieties

Cabbage: Cabaret, Defender, Gladiator, Pusa Muktha

Cauliflower: Pusa ice, Pusa snow ball-K-I-F, Sel-12

Cauliflower mosaic virus (CaMV)

Cauliflower mosaic virus (CaMV)

Cauliflower mosaic virus (CaMV)

Symptoms:

- Vein clearing or banding symptoms are produced on the leaves ultimately leading to mosaic pattern.
- Only the young leaves are affected initially in which light green patches appear.
- Early infection may cause reduction in plant growth leading to stunted look.

Cauliflower mosaic virus (CaMV)

Pathogen:

- The disease is caused by *Cauliflower mosaic virus* which belongs to caulimovirus (CaMV) group.

- Virions are isometric, 50 nm in dia. and non-enveloped and the genome consists of **unipartite ds DNA.**

Cauliflower mosaic virus (CaMV)

Disease cycle and epidemiology:

- The virus is transmitted by mechanical inoculation as well as by different species of aphids particularly *Brevicoryne brassicae* and *Myzus persicae* in semi-persistent manner.

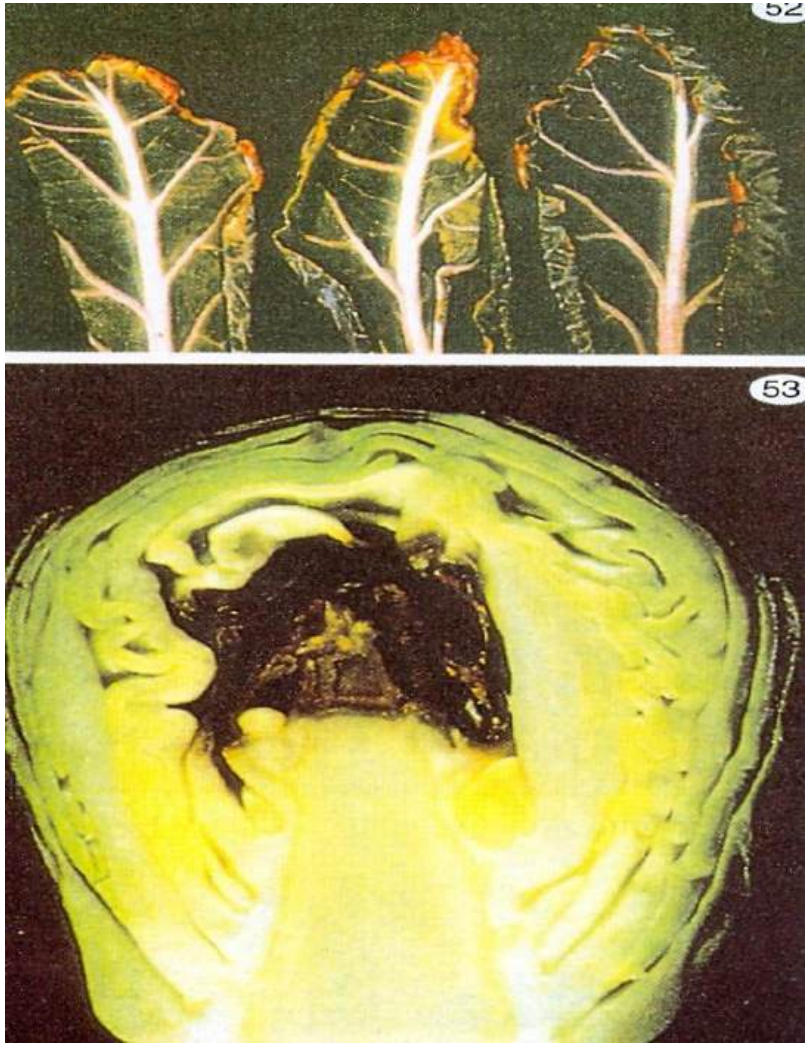
- Besides cabbage and cauliflower, the virus has been reported to infect other Brassica spp. under natural conditions.

Cauliflower mosaic virus (CaMV)

Management:

- Control of aphid vectors through the use of insecticides like Metasystox (0.1%) at an appropriate stage is helpful in reducing the spread of the disease.

Tip burn



- Physiological disorder
- calcium deficiency
- Spray clacium
- Irrigation management
- Avoid excess water

Other diseases of minor importance are:

i. Ring spot : *Mycosphaerella brassicicola* (Fries ex Duby) Lindau

ii. Leaf spots :

Other diseases of minor importance are:

1. Ring spot : *Mycosphaerella brassicicola* (Fries ex Duby) Lindau
2. Leafspots : *Cercospor brassicicola*, *C. cruciferarum*
1. Soft rot or Curd rot : *Erwinia caratovora* subsp. *carotovora*